

17. Information Architecture and User Flows

Canonical information architecture, route system, and user-flow specification

Program	QuantPipeline - private modular portfolio decision platform
Document purpose	Canonical information architecture, route system, and user-flow specification
Status	Working baseline for design and delivery

This document defines the canonical route map, workspace hierarchy, and user flows for QuantPipeline. It is the connective tissue between product scope, UX principles, and the implementation contract used by frontend and backend teams.

1. Information architecture principles

- Organize around decisions, not around model vendors or data-source silos.
- Treat validation and operations as first-class routes, not buried utilities.
- Allow the user to move from summary to evidence to governance without breaking context.
- Preserve a stable shell so route changes feel like focus changes, not product changes.

2. Canonical sitemap

Top-level area	Representative routes	Primary purpose
Overview	/overview, /overview/watch, /overview/regime	triage, monitor, prioritize
Decision	/decision/:id, /decision/:id/challenge, /decision/:id/risk	review and challenge a thesis
Comparison	/compare/:id, /compare/matrix	compare engines and synthesize one stance
Validation	/replay/:snapshotId, /backtests, /paper	inspect historical behavior and paper decisions
Research	/news, /sentiment, /technical, /universe	contextual analysis and input discovery
Admin and ops	/ops, /ops/publication, /ops/incidents, /ops/policies	publication control and reliability

3. Route hierarchy and persistent shell

A route should determine the central canvas and its primary actions, while the surrounding shell remains familiar.

Shell zone	Persistent across routes	Route-specific behavior
Top bar	search, alerts, saved views, global filters	context subtitle and route actions change
Left nav	major sections and current location	sub-navigation adjusts to current top-level area
Center canvas	yes, but content changes by route	main analytical surface
Right context pane	confidence, risk, notes, metadata	content module stack changes by route
Bottom sheet on mobile	global action tray and context	opened context depends on current route

4. Screen inventory

Screen / module	Core objects	Default entry triggers	Must-link destinations
Overview home	recommendation summaries, regime indicators	login, daily revisit	decision, compare, risk
Decision workspace	recommendation object	overview click, alert click	comparison, replay, paper
Comparison matrix	engine outputs, disagreement analysis	decision challenge action	decision, risk, methodology
Risk workspace	portfolio exposures and constraints	risk CTA, breach alert	decision, paper, ops
Replay investigator	snapshot, events, realized outcome	incident or paper miss	ops incident, policy
Backtest lab	strategy runs, assumptions, scorecards	validation work	paper, decision, methodology
Paper portfolio	positions, proposals, drift alerts	monitoring work	decision, replay, ops
Ops command center	queues, health, incidents	operator entry	publication queue, incidents, policies

5. Primary user flows

The flows below are the minimum journeys that the design and QA process must explicitly validate.

Daily review flow

1. Open overview
2. Inspect top recommendation card
3. Enter decision workspace
4. Read evidence and risk summary
5. Save, defer, or move to paper

Required outcome: A recorded decision state with timestamp and notes

Challenge flow

6. Open a recommendation
7. Navigate to comparison
8. Inspect divergence and attribution
9. Change one or more assumptions
10. Return to decision workspace with delta preview

Required outcome: A revised recommendation or an explicitly deferred decision

Replay investigation flow

11. Open replay from incident or paper alert
12. Load historical snapshot
13. Inspect inputs, state, and policy status

- 14. Compare proposed action vs actual result
- 15. Write ops note or create follow-up issue

Required outcome: A resolved investigation artifact

Operator publication flow

- 16. Open ops command center
- 17. Inspect publication queue item
- 18. Check policy and health gates
- 19. Approve, hold, or rollback
- 20. Record explanation and actor

Required outcome: A controlled release action visible in audit history

6. State transitions and route hand-offs

From	To	What must persist	What may reset
Overview	Decision	selected recommendation, timeframe, universe	local sort order on overview widgets
Decision	Comparison	selected recommendation, scenario values	none unless user starts a fresh comparison
Decision	Replay	snapshot key, decision metadata	live-only transient controls
Backtests	Paper	selected strategy assumptions, comparison cohort	temporary chart toggles
Ops queue	Incident detail	item identity, current action rights	list scroll position only

7. Navigation rules and anti-patterns

- Do not create one route per data source if the user expects one decision workspace.
- Do not hide validation and replay behind admin-only patterns; they are part of core trust.
- Do not require the user to reconstruct the current recommendation when moving between comparison and risk.
- Avoid modal overload for critical flows; complex review belongs in full workspaces or drawers.

8. Mobile IA adaptation

Desktop pattern	Mobile equivalent	Reason
Persistent left nav	bottom navigation + more drawer	thumb reach and screen width
Right context rail	bottom sheet / stacked sections	preserve context without crowding
Dense comparison table	accordion rows with key metrics first	legibility
Multi-chart canvas	single primary chart per viewport	focus and performance

9. Flow instrumentation requirements

- Every key flow should emit entry, exit, and decisive action events.
- Instrumentation should distinguish passive viewing from active challenge behavior.
- Abandonment points in challenge and publication flows need explicit event markers.
- Route-level analytics should always include current persona, mode, and scope.

10. IA acceptance criteria

Question	Pass condition
Can the user find the next best action from overview?	yes, within one click
Does challenge preserve context?	yes, recommendation and assumptions persist
Can a replay investigation link back to the original decision?	yes, with stable identifiers
Are operator routes distinct but related to user routes?	yes, shared objects with role-specific actions

QuantPipeline information architecture

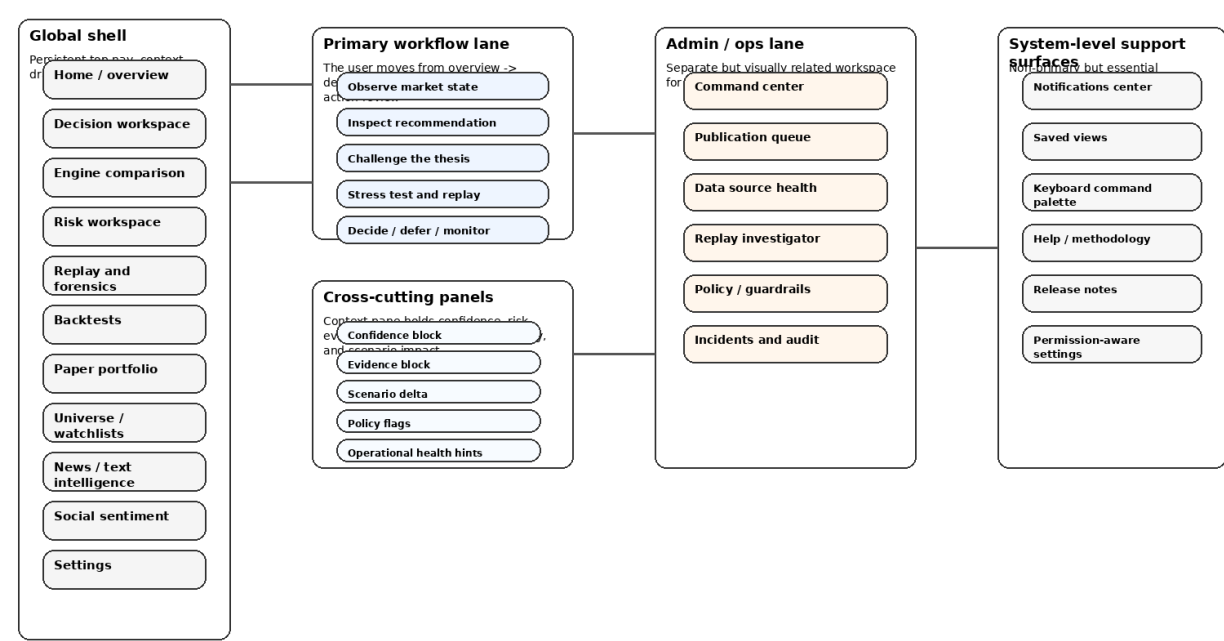


Figure 17-1. Canonical information architecture map.

Key user flows

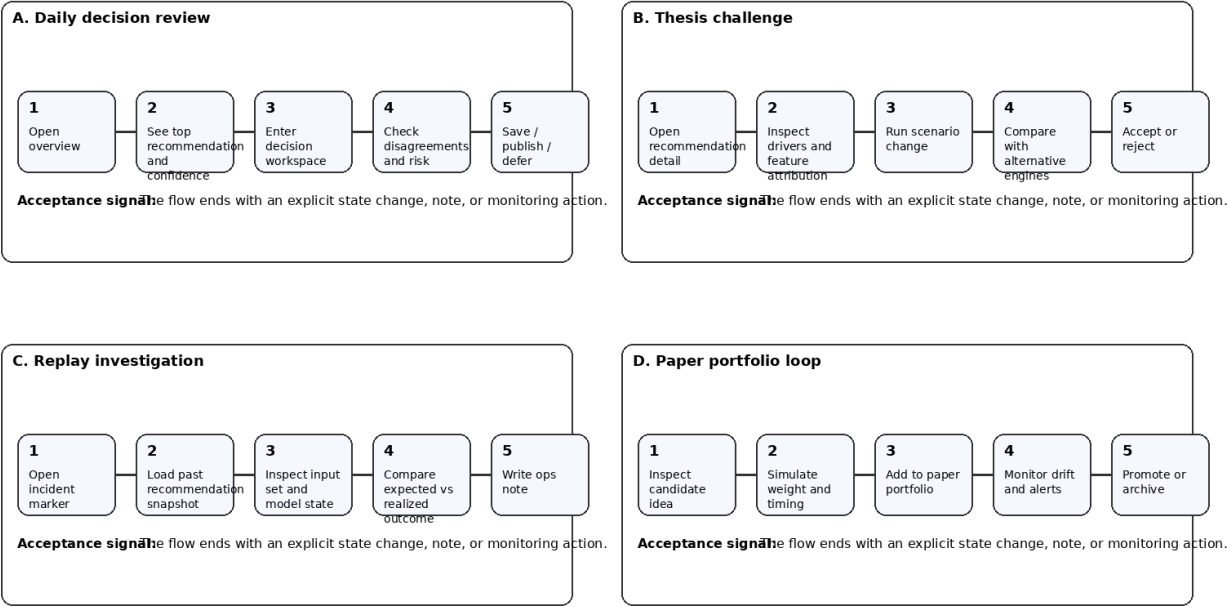


Figure 17-2. Four key flows that the product must support.